Concentration inequalities – Bootcamp

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October 7, 2025

1 Motivation

Concentration inequalities can be seen as "finite-sample versions" of the Law of Large Numbers and the Central Limit Theorem.

- **Law of Large Numbers (LLN).** For i.i.d. random variables $X_1, \ldots, X_n$ with mean μ, the sample mean satisfies
$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{d} \mu, \qquad n \to \infty.$$
Intuition: averages stabilize in the long run.

- **Central Limit Theorem (CLT).** The fluctuations around μ scale as $1/\sqrt{n}$:
$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2).$$
Intuition: in large samples, averages look approximately Gaussian.

- **Motivation for concentration.** Both LLN and CLT are asymptotic, i.e., they describe behavior as $n \to \infty$. Sometimes, though, it is useful to have explicit bounds for finite n, often with small failure probability δ. Concentration inequalities fill this gap:
$$\mathbb{P}\big(\big|\bar{X}_n - \mu\big| \geq \varepsilon\big) \leq \delta,$$
where the scaling of ε depends on n and δ. The nice thing about such bounds is that they are *nonasymptotic*, quantitative, and hold with high probability. With them, we can answer questions like "How many samples are enough for accuracy ε at confidence $1 - \delta$?".

Big picture–why are concentration inequalities useful?

Concentration inequalities quantify how a random quantity (often an average) fluctuates around its mean. They give nonasymptotic, high-probability guarantees such as
$$\mathbb{P}\left(\left|\frac{1}{n}\sum_{i=1}^{n} X_i - \mathbb{E}[X_i]\right| \geq \varepsilon\right) \leq \delta.$$
From such bounds we can find: (i) sample sizes needed for target accuracy ε at confidence $1 - \delta$, (ii) confidence intervals, and (iii) uniform guarantees via union bounds.

2 The basics: Markov and Chebyshev

Theorem 2.1 (Markov's inequality)**.** *Let* $Y \geq 0$ *with* $\mathbb{E}[Y] < \infty$*. For any* $t > 0$*,*

$$\mathbb{P}(Y \geq t) \leq \frac{\mathbb{E}[Y]}{t}.$$

Proof. Since $Y \geq 0$, we have $\mathbb{E}[Y] \geq \mathbb{E}[Y \, \mathbf{1}_{\{Y \geq t\}}] \geq t \, \mathbb{P}(Y \geq t)$. □

Theorem 2.2 (Chebyshev's inequality)**.** *Let* X *have mean* μ *and variance* $\sigma^2 < \infty$*. For any* $t > 0$*,*

$$\mathbb{P}(|X - \mu| \geq t) \leq \frac{\sigma^2}{t^2}.$$

Proof. Apply Markov to $Y = (X - \mu)^2$ with t^2: $\mathbb{P}((X - \mu)^2 \geq t^2) \leq \mathbb{E}[(X - \mu)^2]/t^2 = \sigma^2/t^2$. □

Use

Chebyshev is variance-based and makes minimal assumptions; it is general but often loose. For tighter bounds, we need to make use of the moment generating function.

3 Sub-Gaussian and Sub-Exponential variables

Definition 3.1 (Sub-Gaussian)**.** A mean-zero X is *sub-Gaussian* with variance parameter σ^2 if

$$\mathbb{E}\left[e^{\lambda X}\right] \leq \exp\left(\frac{\sigma^2 \lambda^2}{2}\right),$$

for all $\lambda \in \mathbb{R}$. We write $\|X\|_{\psi_2} \leq C\sigma$, for some absolute C.

Interpretation

Sub-Gaussian variables are those with tails *no heavier* than a Gaussian.

For example, if $X \in [a, b]$ a.s. and $\mathbb{E}X = 0$, then X is sub-Gaussian with $\sigma^2 \leq (b - a)^2/4$. Sub-Gaussian variables guarantee sharp concentration results of the form $\exp(-ct^2)$, which is the basis of Hoeffding's inequality and many finite-sample guarantees in learning theory.

Definition 3.2 (Sub-exponential)**.** A mean-zero X is *sub-exponential* with parameters (ν^2, b) if

$$\mathbb{E}\left[e^{\lambda X}\right] \leq \exp\left(\frac{\nu^2 \lambda^2}{2}\right),$$

for $|\lambda| < 1/b$. We write $\|X\|_{\psi_1} < \infty$. Products of independent sub-Gaussians are sub-exponential.

Interpretation

Sub-exponential variables allow for heavier tails $\exp(-ct)$ (slower than Gaussian decay).

For example, products of independent sub-Gaussians (e.g., $X \cdot Y$ with $X, Y \sim N(0, 1)$ independent) are sub-exponential. Also, the square of a standard Gaussian, $Z^2 - 1$, is sub-exponential but not sub-Gaussian. Sub-Exponential variables naturally arise in variance estimators, χ^2 statistics, and regression residuals.

Example 3.1. Two common examples of sub-Exponential variables are:

1. *Products of independent sub-Gaussians are sub-exponential.*

   Proof. Let X, Y be independent, mean-zero, sub-Gaussian with $\|X\|_{\psi_2} \le K_X$ and $\|Y\|_{\psi_2} \le K_Y$. For $\lambda \in \mathbb{R}$,
$$\mathbb{E}\big[e^{\lambda XY}\big] = \mathbb{E}\Big[\mathbb{E}\big[e^{\lambda XY} \mid X\big]\Big] \;\le\; \mathbb{E}\Big[\exp\Big(\tfrac{\lambda^2 X^2 K_Y^2}{2}\Big)\Big],$$
   using sub-Gaussianity of Y. Now, by the standard sub-Gaussian moment bound, $\mathbb{E}\big[e^{cX^2}\big] \le 2$ provided $c \lesssim K_X^{-2}$. Hence, for $|\lambda| \le \frac{c_0}{K_X K_Y}$, we have
$$\mathbb{E}\Big[\exp\Big(\tfrac{\lambda^2 X^2 K_Y^2}{2}\Big)\Big] \;\le\; 2 \;\le\; \exp\big(C\,\lambda^2 K_X^2 K_Y^2\big),$$
   for absolute constants $c_0, C > 0$. Therefore,
$$\mathbb{E}\big[e^{\lambda XY}\big] \;\le\; \exp\Big(\tfrac{\nu^2\lambda^2}{2}\Big) \quad \text{for } |\lambda| < 1/b,$$
   with parameters $\nu^2 \asymp K_X^2 K_Y^2$ and $b \asymp K_X K_Y$. This is exactly the sub-exponential mgf condition. □

2. *If $Z \sim N(0,1)$, then $Z^2 - 1$ is sub-exponential but not sub-Gaussian.*

   Proof. Let $Z \sim N(0,1)$ and $Y := Z^2 - 1$. The mgf of Z^2 is $\mathbb{E}[e^{tZ^2}] = (1-2t)^{-1/2}$ for $t < \frac{1}{2}$. Hence, for $|\lambda| < \frac{1}{2}$,
$$\mathbb{E}\big[e^{\lambda Y}\big] = \mathbb{E}\big[e^{\lambda(Z^2-1)}\big] = e^{-\lambda}(1-2\lambda)^{-1/2}.$$
   A Taylor expansion at $\lambda = 0$ gives
$$\log \mathbb{E}\big[e^{\lambda Y}\big] = -\lambda - \tfrac{1}{2}\log(1-2\lambda) = \lambda^2 + O(\lambda^3),$$
   where we used $\log(1-t) = -t - \frac{t^2}{2} - \frac{t^3}{3} - \dots$, with $t = 2\lambda$. Thus, for sufficiently small $|\lambda|$ we have $\mathbb{E}[e^{\lambda Y}] \le \exp(C\lambda^2)$, i.e., Y is sub-exponential. Note, however, that Y is *not* sub-Gaussian: its mgf $\mathbb{E}[e^{\lambda Y}]$ is finite only for $\lambda < \frac{1}{2}$, whereas a sub-Gaussian X satisfies $\mathbb{E}[e^{\lambda X}] \le \exp(\sigma^2\lambda^2/2)$ for *all* $\lambda \in \mathbb{R}$. □

Applications in statistics

- *High-dimensional regression:* To prove consistency of Lasso or Ridge, we need sub-Gaussian columns of the design matrix X to control deviations of sample covariances.
- *PCA and covariance estimation:* Sample covariance matrices concentrate sharply around their expectation under sub-Gaussian assumptions.
- *Likelihood ratios and generalized linear models:* Products of covariates and errors are often only sub-exponential, not sub-Gaussian — motivating Bernstein's inequality.

4 Exponential inequalities

Theorem 4.1 (Hoeffding's inequality)**.** *Let $X_1, \dots, X_n$ be independent with $X_i \in [a_i, b_i]$ a.s. and $\mathbb{E}X_i = \mu_i$. For $S_n = \sum_{i=1}^n X_i$ and any $t > 0$,*

$$\mathbb{P}\big(S_n - \mathbb{E}S_n \geq t\big) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right),$$

and the same bound holds for the lower tail; by symmetry,

$$\mathbb{P}\left(\left|\frac{1}{n}\sum_{i=1}^n X_i - \frac{1}{n}\sum_{i=1}^n \mu_i\right| \geq \varepsilon\right) \leq 2\exp\left(-\frac{2n^2\varepsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

Theorem 4.2 (Chernoff bound for Bernoulli sums)**.** *Let $X_i \sim \mathrm{Bernoulli}(p)$ independent, $S_n = \sum_{i=1}^n X_i$, and $\bar{X} = S_n/n$. For any $\varepsilon \in (0, 1-p]$,*

$$\mathbb{P}\big(\bar{X} \geq p + \varepsilon\big) \leq \exp\big(-n\, D_{\mathrm{KL}}(p + \varepsilon \,\|\, p)\big),$$

where $D_{\mathrm{KL}}(q\|p) = q\log\frac{q}{p} + (1-q)\log\frac{1-q}{1-p}$. A convenient form is

$$\mathbb{P}\big(|\bar{X} - p| \geq \varepsilon\big) \leq 2\exp(-2n\varepsilon^2) \quad \textit{(Hoeffding form).}$$

Chernoff's idea

Chernoff found a nice and powerful way of using Markov's inequality: for any random variable S and any $\lambda > 0$, we have

$$\mathbb{P}(S \geq t) = \mathbb{P}\left(e^{\lambda S} \geq e^{\lambda t}\right) \leq e^{-\lambda t}\mathbb{E}[e^{\lambda S}] = \exp\left(-\lambda t + \log\mathbb{E}[e^{\lambda S}]\right).$$

Since this holds **for any** $\lambda > 0$, we can *optimize* using simple Calculus and get

$$\mathbb{P}(S \geq t) \leq \inf_{\lambda > 0} \exp\left(-\lambda t + \log\mathbb{E}[e^{\lambda S}]\right).$$

This is how the proofs of several famous concentration inequalities go: (i) bound the mgf; (ii) use Chernoff's trick.

Theorem 4.3 (Bernstein's inequality)**.** *Let X_i be independent, mean-zero, $|X_i| \leq M$ a.s., and $\mathrm{Var}(X_i) \leq \sigma_i^2$. For $S_n = \sum_{i=1}^n X_i$ and any $t > 0$,*

$$\mathbb{P}\big(S_n \geq t\big) \leq \exp\left(-\frac{t^2}{2\sum_{i=1}^n \sigma_i^2 + \frac{2}{3}Mt}\right).$$

Equivalently, for the average $\bar{X} = \frac{1}{n}\sum X_i$ with common variance σ^2 and bound $|X_i| \leq M$,

$$\mathbb{P}\big(|\bar{X} - \mathbb{E}\bar{X}| \geq \varepsilon\big) \leq 2\exp\left(-\frac{n\varepsilon^2}{2\sigma^2 + \frac{2}{3}M\varepsilon}\right).$$

Hoeffding vs. Bernstein

Hoeffding uses only boundedness (no variance), while Bernstein adapts to the variance for small deviations and transitions to Hoeffding-like decay for larger ε.

Theorem 4.4 (McDiarmid's inequality (bounded differences))**.** *Let $X_1, \dots, X_n$ be independent, and f satisfy for each i and all inputs,*

$$\sup_{x_1,\dots,x_n,x_i'} \big| f(x_1,\dots,x_i,\dots,x_n) - f(x_1,\dots,x_i',\dots,x_n) \big| \le c_i.$$

Then, for any $t > 0$,

$$\mathbb{P}\big(f(X) - \mathbb{E}f(X) \ge t\big) \le \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right),$$

and likewise for the lower tail.

Theorem 4.5 (Hanson–Wright inequality)**.** *Let $X = (X_1, \dots, X_n)^\top$ have independent, mean-zero, sub-Gaussian coordinates with*

$$\|X_i\|_{\psi_2} \;\le\; K \qquad (1 \le i \le n).$$

Let $A \in \mathbb{R}^{n\times n}$ be a fixed matrix, and write $\|A\|_{op}$ for the operator norm and $\|A\|_F$ for the Frobenius norm. Then for every $t > 0$,

$$\mathbb{P}\left(\Big| X^\top A X - \mathbb{E}[X^\top A X] \Big| \ge t\right) \;\le\; 2\exp\left[-c\,\min\left(\frac{t^2}{K^4\,\|A\|_F^2},\, \frac{t}{K^2\,\|A\|_{op}}\right)\right],$$

where $c > 0$ is an absolute constant.

Intuition

Quadratic forms of independent sub-Gaussian vectors concentrate sharply around their mean.

- For *moderate* deviations ($t \lesssim K^2\|A\|_F^2/\|A\|_{op}$), the tail is $\exp\big(-c\,t^2/(K^4\|A\|_F^2)\big)$ — a sub-Gaussian regime governed by the "size" $\|A\|_F$.
- For *large* deviations, the tail becomes $\exp\big(-c\,t/(K^2\|A\|_{op})\big)$ — a sub-exponential regime governed by the "worst direction" $\|A\|_{op}$.

This is the natural analogue of Bernstein-type bounds for quadratic (rather than linear) functionals.

5 Statistical applications

(A) Confidence intervals for means (bounded/sub-Gaussian data)

Let $X_1, \dots, X_n$ be i.i.d. with mean μ.

a) **Hoeffding (bounded data).** If $X_i \in [a, b]$ a.s., then for any $\delta \in (0, 1)$,

$$\mathbb{P}\left(\big|\bar{X} - \mu\big| \le \sqrt{\frac{(b-a)^2}{2n}\,\log\frac{2}{\delta}}\right) \ge 1 - \delta.$$

To see why this is the case, note that if $X_i \in [a, b]$ a.s., then $X_i - \mu$ is sub-Gaussian with parameter $(b-a)^2/4$, by Hoeffding's lemma. Thus, by Chernoff's method,

$$\mathbb{P}\big(\bar{X} - \mu \ge t\big) \le \exp\left(-\frac{2nt^2}{(b-a)^2}\right).$$

Symmetry gives the two-sided bound, so with probability $1 - \delta$,

$$|\bar{X} - \mu| \le \sqrt{\frac{(b-a)^2}{2n}\,\log\frac{2}{\delta}}.$$

b) **Sub-Gaussian.** If $X_i - \mu$ is sub-Gaussian with parameter σ^2, then

$$\mathbb{P}\left(|\bar{X} - \mu| \le \sigma\sqrt{\frac{2}{n}\log\frac{2}{\delta}}\right) \ge 1 - \delta.$$

To see why, note that if $X_i - \mu$ is sub-Gaussian(σ^2), then for any λ,

$$\mathbb{E}[e^{\lambda(\bar{X}-\mu)}] = \prod_{i=1}^{n} \mathbb{E}[e^{\lambda(X_i-\mu)/n}] \le \exp\left(\frac{\sigma^2\lambda^2}{2n}\right).$$

Thus $\bar{X} - \mu$ is itself sub-Gaussian with variance proxy σ^2/n, and

$$\mathbb{P}\big(|\bar{X} - \mu| \ge t\big) \le 2\exp\left(-\frac{nt^2}{2\sigma^2}\right).$$

Choosing $t = \sigma\sqrt{\frac{2}{n}\log\frac{2}{\delta}}$ gives the stated confidence bound.

Sample complexity

If we want accuracy ε with confidence $1 - \delta$, we need sample size

$$n \gtrsim \begin{cases} \frac{(b-a)^2}{2\varepsilon^2}\log\frac{2}{\delta} & \text{(bounded)} \\ \frac{2\sigma^2}{\varepsilon^2}\log\frac{2}{\delta} & \text{(sub-Gaussian).} \end{cases}$$

(B) Estimating a coin bias (Chernoff/Hoeffding)

For $X_i \sim \text{Bernoulli}(p)$ i.i.d., note $X_i \in [0, 1]$ so Hoeffding applies with $b - a = 1$. Hence,

$$\mathbb{P}(|\bar{X} - p| \ge t) \le 2\exp(-2nt^2).$$

Setting $t = \sqrt{\frac{1}{2n}\log\frac{2}{\delta}}$ gives

$$\mathbb{P}\left(|\bar{X} - p| \le \sqrt{\frac{1}{2n}\log\frac{2}{\delta}}\right) \ge 1 - \delta.$$

Equivalently, to guarantee accuracy ε, solve

$$2\exp(-2n\varepsilon^2) \le \delta \quad \Longrightarrow \quad n \ge \frac{1}{2\varepsilon^2}\log\frac{2}{\delta}.$$

(C) Multiple comparisons via union bound

Suppose we estimate means μ_j of d features using $\bar{X}^{(j)}$ from independent (or even common-sample) bounded observations. For each coordinate j, Hoeffding gives

$$\mathbb{P}\left(|\bar{X}^{(j)} - \mu_j| \ge t\right) \le 2\exp\left(-\frac{2nt^2}{(b-a)^2}\right).$$

By the union bound,

$$\mathbb{P}\left(\max_{1\le j\le d} |\bar{X}^{(j)} - \mu_j| \ge t\right) \le 2d\exp\left(-\frac{2nt^2}{(b-a)^2}\right).$$

Rearranging for δ yields

$$\mathbb{P}\left(\max_{1\le j\le d} |\bar{X}^{(j)} - \mu_j| \le \sqrt{\frac{(b-a)^2}{2n}\log\frac{2d}{\delta}}\right) \ge 1-\delta.$$

High dimensions

Uniform control introduces a $\log d$ factor. This is the basic path for high-dimensional guarantees and simultaneous confidence bands.

(D) Empirical risk minimization (bounded losses)

Let $\ell(\theta; Z) \in [0,1]$ and $R(\theta) = \mathbb{E}\,\ell(\theta; Z)$, $\hat{R}_n(\theta) = \frac{1}{n}\sum_{i=1}^n \ell(\theta; Z_i)$. For a *finite* hypothesis set Θ with $|\Theta| = N$, Hoeffding yields, for any $\delta \in (0,1)$,

$$\mathbb{P}(|\hat{R}_n(\theta) - R(\theta)| \ge t) \le 2\exp(-2nt^2),$$

for fixed θ. Now apply the union bound over $N = |\Theta|$:

$$\mathbb{P}\left(\sup_{\theta\in\Theta} |\hat{R}_n(\theta) - R(\theta)| \ge t\right) \le 2N\exp(-2nt^2).$$

Set $t = \sqrt{\frac{1}{2n}\log\frac{2N}{\delta}}$ to ensure the RHS $\le \delta$, giving

$$\mathbb{P}\left(\sup_{\theta\in\Theta} |\hat{R}_n(\theta) - R(\theta)| \le \sqrt{\frac{1}{2n}\log\frac{2N}{\delta}}\right) \ge 1-\delta.$$

This gives a uniform generalization bound.

6 Choosing a tool

- **Chebyshev**: minimal assumptions, variance-only; loose but universal.
- **Hoeffding**: bounded variables; simple, dimension-free, great for means of bounded/Bernoulli data.
- **Bernstein**: exploits variance and boundedness; tighter for small deviations.
- **McDiarmid**: concentration of Lipschitz functions of independent inputs (e.g., U-statistics).

Remark 6.1 (From tails to confidence intervals)**.** A tail bound of the form $\mathbb{P}(|\hat{\theta} - \theta| \ge \varepsilon) \le \delta(\varepsilon, n)$ immediately yields a $(1-\delta)$ CI by solving for ε; we can then determine the required sample size by solving for n.

6.1 Proof sketches

Theorem 6.1 (Markov's inequality)**.** *Let $Y \ge 0$ with $\mathbb{E}[Y] < \infty$. For any $t > 0$,*

$$\mathbb{P}(Y \ge t) \le \frac{\mathbb{E}[Y]}{t}.$$

Proof. We observe that $\mathbb{E}[Y] \ge \mathbb{E}[Y\,\mathbf{1}_{\{Y\ge t\}}] \ge t\,\mathbb{P}(Y \ge t)$, hence the claim. □

Theorem 6.2 (Chebyshev's inequality)**.** *If X has mean μ and variance $\sigma^2 < \infty$, then for $t > 0$,*

$$\mathbb{P}(|X-\mu| \geq t) \leq \frac{\sigma^2}{t^2}.$$

Proof. Apply Markov to $Y = (X-\mu)^2$ with threshold t^2. □

Lemma 6.1 (Hoeffding's Lemma)**.** *Let Z be a mean-zero random variable with $Z \in [a,b]$ a.s. Then for all $\lambda \in \mathbb{R}$,*

$$\mathbb{E}\big[e^{\lambda Z}\big] \;\leq\; \exp\!\left(\frac{\lambda^2(b-a)^2}{8}\right).$$

Proof sketch. By convexity of $e^{\lambda z}$, for $z \in [a,b]$, $e^{\lambda z} \leq \frac{b-z}{b-a}e^{\lambda a} + \frac{z-a}{b-a}e^{\lambda b}$. Taking expectation and using $\mathbb{E}Z = 0$ gives an upper bound that simplifies to the stated sub-Gaussian mgf with variance proxy $(b-a)^2/4$. Details follow from algebra and the inequality $\cosh(u) \leq e^{u^2/2}$. □

Theorem 6.3 (Hoeffding's inequality)**.** *Let $X_1, \ldots, X_n$ be independent with $X_i \in [a_i, b_i]$ a.s. and $\mu_i = \mathbb{E}X_i$. Put $S_n = \sum_{i=1}^n (X_i - \mu_i)$. Then for any $t > 0$,*

$$\mathbb{P}(S_n \geq t) \leq \exp\!\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right),$$

and the same bound holds for the lower tail.

Proof. For any $\lambda > 0$, by independence and Hoeffding's Lemma,

$$\mathbb{E}e^{\lambda S_n} = \prod_{i=1}^n \mathbb{E}e^{\lambda(X_i-\mu_i)} \leq \exp\!\left(\frac{\lambda^2}{8}\sum_{i=1}^n (b_i-a_i)^2\right).$$

By Markov's inequality,

$$\mathbb{P}(S_n \geq t) = \mathbb{P}\left(e^{\lambda S_n} \geq e^{\lambda t}\right) \leq e^{-\lambda t}\,\mathbb{E}e^{\lambda S_n} \leq \exp\!\left(-\lambda t + \frac{\lambda^2}{8}\sum_{i=1}^n (b_i-a_i)^2\right).$$

Optimizing in λ gives $\lambda^\star = \frac{4t}{\sum (b_i - a_i)^2}$ and the stated bound. □

Theorem 6.4 (Bernstein's inequality)**.** *Let $X_1, \ldots, X_n$ be independent, mean-zero, $|X_i| \leq M$ a.s., and $\mathrm{Var}(X_i) \leq \sigma_i^2$. Put $S_n = \sum_{i=1}^n X_i$ and $v = \sum_{i=1}^n \sigma_i^2$. Then for all $t > 0$,*

$$\mathbb{P}(S_n \geq t) \leq \exp\!\left(-\frac{t^2}{2v + \frac{2}{3}Mt}\right).$$

Proof sketch. A standard mgf bound for bounded mean-zero X (via $e^u \leq 1 + u + \frac{u^2}{2(1-u/3)}$ for $u \in [0,3)$) yields

$$\log \mathbb{E}e^{\lambda X_i} \leq \frac{\lambda^2\,\mathbb{E}[X_i^2]}{2(1-\lambda M/3)} \quad (|\lambda| < 3/M).$$

By independence, $\log \mathbb{E}e^{\lambda S_n} \leq \frac{\lambda^2 v}{2(1-\lambda M/3)}$. Chernoff + optimize in $\lambda \in (0, 3/M)$ yields the bound. □

Theorem 6.5 (Chernoff bound for Bernoulli sums)**.** *Let $X_i \sim \mathrm{Bernoulli}(p)$ i.i.d., $S_n = \sum_{i=1}^n X_i$, and $q \in (p,1)$. Then,*

$$\mathbb{P}\!\left(\frac{S_n}{n} \geq q\right) \leq \exp\big(-n\,D_{\mathrm{KL}}(q\|p)\big),$$

where $D_{\mathrm{KL}}(q\|p) = q\log\frac{q}{p} + (1-q)\log\frac{1-q}{1-p}$.

Proof. For $\lambda > 0$, $\mathbb{E}e^{\lambda X_i} = pe^{\lambda} + (1-p)$. Independence gives $\mathbb{E}e^{\lambda S_n} = (pe^{\lambda} + 1 - p)^n$. By Chernoff,

$$\mathbb{P}(S_n \geq nq) \leq \inf_{\lambda>0} \exp\big(-\lambda nq\big)\,(pe^{\lambda} + 1 - p)^n.$$

Minimizing the rhs occurs at $e^{\lambda^\star} = \frac{q(1-p)}{p(1-q)}$, and plugging back yields the KL form. □

Theorem 6.6 (McDiarmid's inequality (bounded differences))**.** *Let $X_1, \dots, X_n$ be independent, and f satisfy for each i,*

$$\sup_{x_1,\dots,x_n,x_i'} \big|f(x_1,\dots,x_i,\dots,x_n) - f(x_1,\dots,x_i',\dots,x_n)\big| \leq c_i.$$

Then, for all $t > 0$,

$$\mathbb{P}\big(f(X) - \mathbb{E}f(X) \geq t\big) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right).$$

Proof sketch. The sequence of random variables $Y_i = \mathbb{E}[f(X) \mid X_1, \dots, X_i]$ defines a martingale. Then, $Y_0 = \mathbb{E}f(X)$, $Y_n = f(X)$, and the martingale differences $D_i = Y_i - Y_{i-1}$ satisfy $|D_i| \leq c_i/2$ a.s., by the bounded-differences property. The Azuma–Hoeffding inequality for martingales gives

$$\mathbb{P}\left(\sum_{i=1}^n D_i \geq t\right) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right).$$

□

Lemma 6.2 (Union bound)**.** *For events $E_1, \dots, E_d$,*

$$\mathbb{P}\Big(\bigcup_{j=1}^d E_j\Big) \leq \sum_{j=1}^d \mathbb{P}(E_j).$$

Proof. We have $\mathbf{1}_{\cup_j E_j} \leq \sum_j \mathbf{1}_{E_j}$ pointwise. Take expectations and use that $\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A)$. □